

Multimodal AI-Based Phishing Detection Using Email Text, Webpage Screenshots, and URL Features

Group 07
Section 01

Abstract—A phishing attack can come in the form of a fake email, fake webpage, or suspicious link. The objective of this project is to build an information light multimodal phishing detection system based on these three information. In the first case, the text features in the email are represented by TF-IDF and classified by Logistic Regression, in the second case, the features extracted from the screenshots of the webpages are extracted using the frozen MobileNetV2 and then classified by Logistic Regression, in the third case, the features of the URL are classified by Random Forest. They predict probabilities of the events and these probabilities are combined by late fusion. The email, URL and screenshot experiments got F1-scores of 0.9905, 0.9615 and 0.6400 respectively on the main test sets; the fused experiment had F1 on the main test sets of 0.9892. Explainability, feature ablation and cross-dataset validation were also done.

Index Terms—phishing detection, multimodal learning, TF-IDF, Random Forest, MobileNetV2, late fusion, explainable AI

I. INTRODUCTION

Often a phishing attack will rely on several types of deception. The URL might have unexpected attributes, the link might redirect to a fake site and an email could include a malicious link. This project investigates these three modalities independently and then merges the predictions of these three modalities together. The goal is to create a system that is both practical and CPU efficient, with the use of simple machine-learning techniques.

II. DATASET

The following public datasets were used. For the email experiment, the CEAS_08 was used along with a balanced subset of 9,000 emails, with 4,500 legitimate and 4,500 phishing emails. After preprocessing the URL dataset had 11,430 records and 87 numerical features. The data set used for the screenshots was 1697 images, of which 1150 was a balanced data set containing 600 real screenshots and 550 phishing screenshots. Split of the data into stratified train, validation and test sets.

III. METHODOLOGY

A. Email Classification

The text in the subject line and body of the emails was cleaned and transformed to TF-IDF features with the top 20,000 features, which included unigram and bigram. Logistic Regression was chosen because it is a method that has good performance and is easy to compute.

B. Screenshot Classification

The images shown were scaled down to 160x160. For these features, a pretrained MobileNetV2 was retained, and used as a feature extractor, generating 1280 dimensional representations. These features were then used to train Logistic Regression. On the main subset, the notebook records the time of about 25.7 seconds to extract features.

C. URL Classification

The URL dataset contained features extracted beforehand, which were numerical. Logistic Regression and Random Forest were compared and the result of validation was better with Random Forest. Later, hyper parameter tuning was done via GridSearchCV.

D. Late Fusion

There is no common identifier for the attacks in the three public datasets, meaning exact email–screenshot–URL pairs are not available. After the project proposal, the training of each modality was done independently and the probabilities were added together. A synthetic class matched validation/test set was thus generated. The weights were chosen with the help of validation.

$$p_F = 0.5p_E + 0.3p_S + 0.2p_U. \quad (1)$$

This synthetic evaluation is a limitation and is discussed later.

IV. LITERATURE REVIEW

There are three studies identified in the project proposal. To support the use of structured webpage and URL evidence, Li *et al.* provided two features (URL and HTML features) and a stacking model for detecting phishing webpage. Al-Subaiey *et al.* designed an explainable phishing-email detection system that was interpretable, which prompted text-based phishing-email detection and explainability. The multimodal probability-fusion idea employed here is directly inspired by the concept of a multimodal phishing website system made by Vulfin *et al.* that makes use of explainable AI and late fusion. In the present work, simpler CPU-friendly models are used and all the evidence, such as email, screenshot and URL, have been combined.

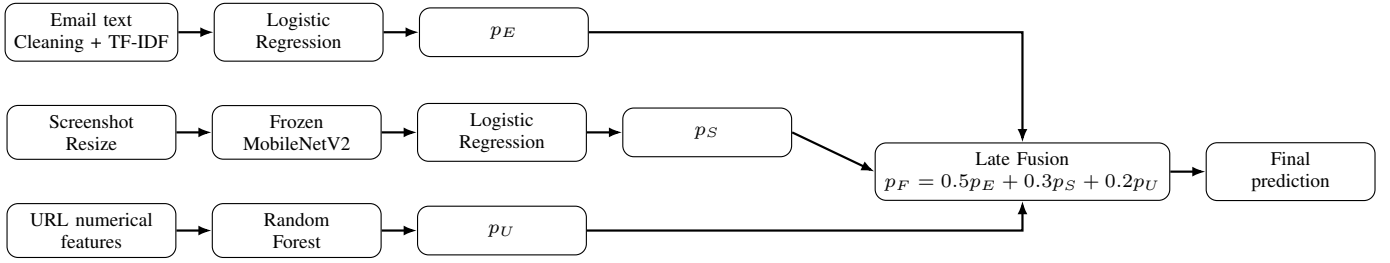


Fig. 1. Multimodal phishing detection pipeline.

TABLE I
MAIN TEST-SET RESULTS

Model	Acc.	Prec.	Rec.	F1	AUC
Email-only	.9906	.9944	.9867	.9905	.9996
URL-only	.9615	.9615	.9615	.9615	.9929
Screenshot-only	.6478	.6261	.6545	.6400	.6553
Combined fusion	.9891	.9828	.9957	.9892	.9991

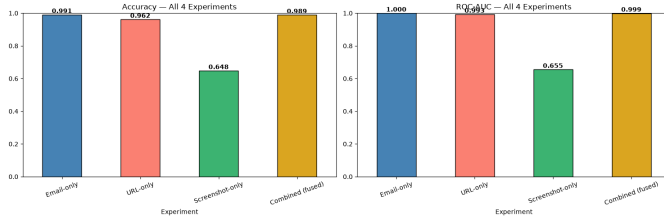


Fig. 2. Accuracy and ROC-AUC comparison from the notebook.

V. RESULTS AND ANALYSIS

Table I presents the main test results.

The best single modality was the email. The classification of URL's was also good and the classification of screenshots was significantly more challenging. The fused experiment got the recall of 99.57% for phishing and F1 score of 98.92%. This effect is not to be interpreted as a result on true paired multimodal attacks as these were synthetic class-matched combinations.

A. Hyperparameter Tuning

The upgrades to all three of the modality models were improved by tuning. The tuned URL Random Forest achieved an accuracy of 0.9650, an F1 score of 0.9651 and an AUC of 0.9944 with $n_estimators = 300$ and the maximum depth of 20 and minimum leaf size of 1. The tuned email Logistic Regression used $C = 10$ and reached 0.9917 accuracy, 0.9916 F1, and 0.9997 AUC. The accuracy for tuned screenshot Logistic Regression was 0.6652 with F1 of 0.6751 and AUC of 0.6808.

B. Feature Analysis

The most important URL features included *google_index*, *page_rank*, *nb_hyperlinks*, *web_traffic*, and *nb_www*. Removing *page_rank* produced the largest accuracy drop among the tested top features (0.0066), while some feature removals caused little or no drop, suggesting redundancy.

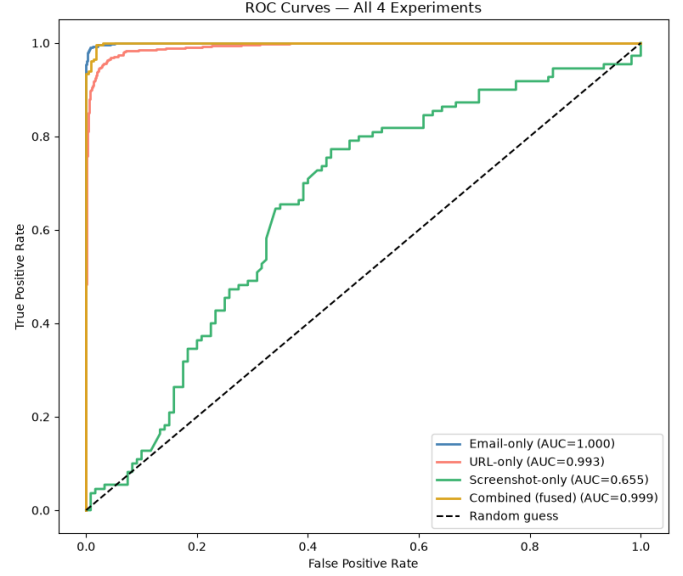


Fig. 3. ROC curves for the four main experiments.

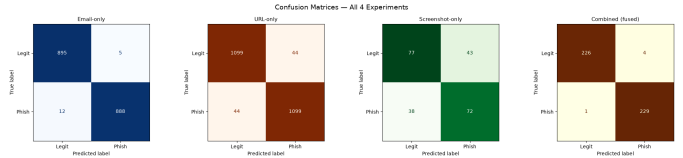


Fig. 4. Confusion matrices for the four experiments.

C. Explainability

The project compute TF-IDF term importance for email classification, Grad-CAM visualizations of screenshots and explanations of URL features using Shapley values. They will include evidence from the text, URL, and/or visual information that affected a prediction.

VI. EXTERNAL VALIDATION

The project also evaluated the models on data sets that they hadn't been trained on themselves. Email accuracy and ROC-AUC decreased from 0.9906/0.9996 on CEAS_08 to 0.7309/0.7758 on Enron. With only nine features, this basic model outperformed on the original test set (0.8325/0.9148), and on PhiUSIIL (0.7911/0.9453). The external Phish-IRIS subset found to be 0.5417/0.6469 for the screenshot model.

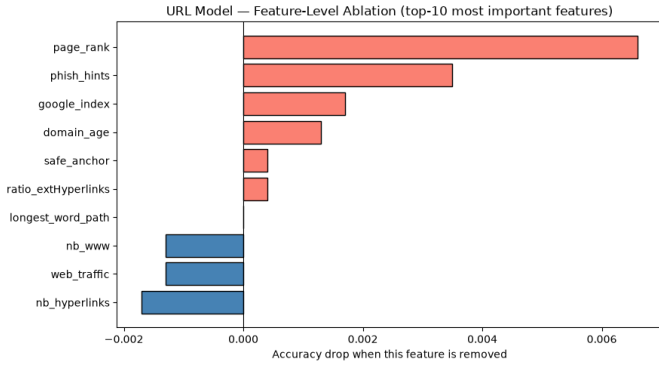


Fig. 5. URL feature-level ablation study.

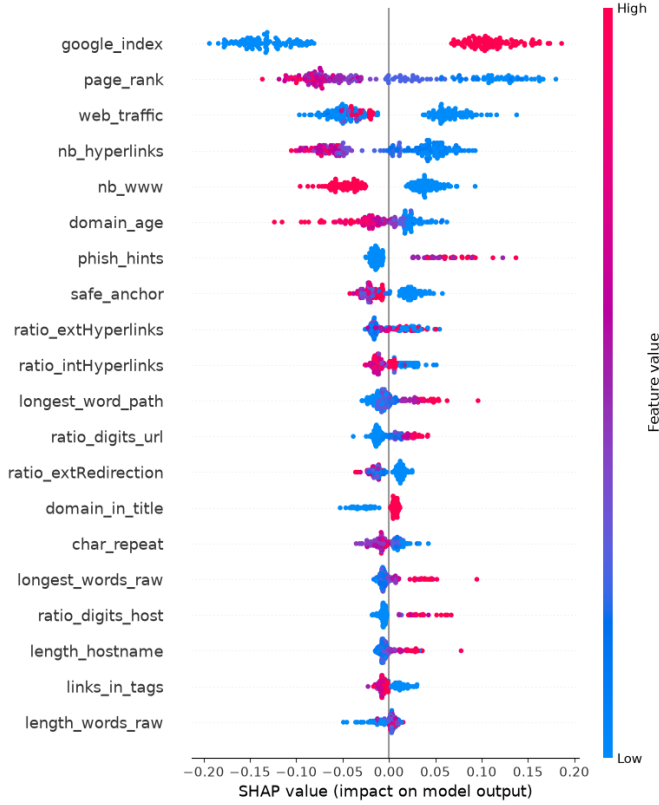


Fig. 6. SHAP summary for URL features.

The differences highlight the distribution shift and reveal that there is no guarantee that the external performance of the model is equivalent if the in-dataset performance is good.

VII. ERROR ANALYSIS

Examples of misclassified email were short legitimate emails and promotional phishing emails. When the URL characteristics were in conflict, there were errors. The screenshot model proved to be more challenging due to the fact that the visual features of legitimate and phishing web pages can be similar. These observations correspond to the low level of screenshots test results.

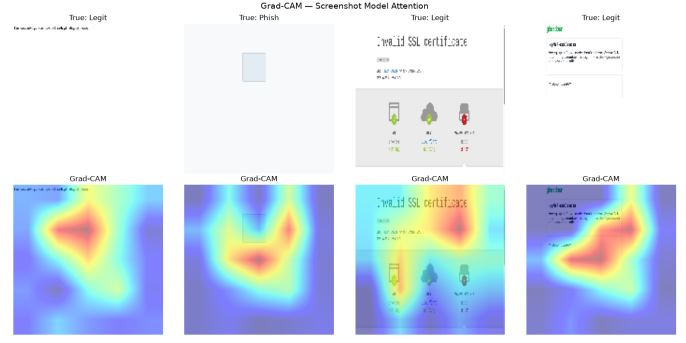


Fig. 7. Grad-CAM examples for screenshot predictions.

VIII. LIMITATIONS

The primary drawback is that common identifiers of attacks across the three public datasets are not available. Therefore, it was impossible to conduct a fusion experiment with true pairs of multimodal samples. Because the screen shot data set is not very large and distribution shift is confirmed by external assessment. The system also circumvents and runs live URLs.

IX. FUTURE WORK AND CONCLUSION

In future studies, genuine paired multimodal phishing data is needed, a larger set of screenshots, a more powerful language model, a vision model, and dynamic fusion weights. Temporal and Adversarial validation could take this further to test robustness to novel attacks. The overall project is a practical light weight multimodal pipeline. The results were generally good for the models that used emails and URLs, and the primary needs for improvement are in the area of screenshot classification.

REFERENCES

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